



VANN BROTHERS

Functional Design Document

AgriErp

Variance Calculation

Revision History

Version No.	Date	Description	Author
1.0	June 9, 2026	Variance Calculation	Kamil
1.1	June 11, 2026	Variance Calculation	Shuraim

1. Introduction

1.1. Purpose

The purpose of this customization is to create a batch process to compare labor amounts posted through General Ledger journals against labor costs posted from AgriERP Expense Journals on Projects

The system will calculate the variance by Plot/Field and post the variance through an expense journal on projects.

1.2. Scope

- Monthly batch job for GL vs Project labor variance calculation.
- Comparison of GL transactions posted on accounts Labor Absorption accounts.
- Comparison of AgriERP labor costs posted through Project Expense Journals.
- Variance calculation by Plot/Field.
- Automatic Expense Journal creation and posting on projects.
- Setup form for debit and credit posting accounts.

1.3. Objective

The objective is to automate month-end labor variance posting between GL labor postings and AgriERP labor postings, while ensuring the variance is recorded against the correct Plot/Field.

2. Functional Requirements

This section defines the requirements for the GL vs Project Labor Variance Posting customization

2.1. Menu Item Placement

- **Batch Job**

AgriERP Management > Periodic Task > AgriERP Labor Variance Posting

- **Setup Form**

AgriERP Management > Setup > AgriERP Labor Variance Setup

- **New Drop Down Field**

A new drop down field is created in AgriERP Setup to set the absorption account as offset in the GL.

2.2. Batch Job Requirement

A new batch job named AgriERP Labor Variance Posting will be created.

Parameters

Parameter	Type	Mandatory	Notes
From Date	Date	Yes	Start date for transaction selection.
To Date	Date	Yes	End date for transaction selection.
Posting Date	Date	Yes	Date used for variance journal posting.
Journal Name	Lookup	Yes	General Journal name used for posting.
Batch Processing	Yes/No	No	Allows the process to run in batch.

2.3. Absorption Account field

A new drop-down field will be created in the AgriERP parameter to define the accounts used for variance posting.

Field	Type	Data	Editable
Absorption Account	Dropdown	Main Account	Yes

2.4. Variance Calculation Logic

The batch job will calculate the variance at the Plot/Field level.

1. GL Transaction Selection

The system will select posted GL transactions where:

- Posting date is within the selected date range.
- Main Account is Labor Absorption account.
- Plot/Field financial dimension is populated.

2. Project Expense Transaction Selection

The system will select posted Project Expense Journal transactions created from AgriERP where:

- Posting date is within the selected date range.
- Project is linked to a Plot/Field in master data.

The system will derive Plot/Field from the Project master and aggregate the labor cost amount by Plot/Field.

3. Variance Formula

$$\text{Variance} = \text{Labor Absorption GL Account} - \text{Project Expense Amount}$$

Only non-zero variances will be posted.

4. Posting Logic

If Variance Amount is Positive:

- Amount will be on the debit column of the Expense Journal Line

If Variance Amount is Negative:

- Amount will be on the Credit column of the Expense Journal Line

If Variance Amount is Zero:

- No journal line will be created.

2.5. Integration Behavior

- The batch job will be run manually at month-end.
- The system will compare GL and Project Expense amounts by Plot/Field.
- The system will create one variance journal entry per Plot/Field where variance exists.
- The variance journal must carry all the related financial dimensions available in the GL account, i.e., Labor Absorption account.
- The journal must be validated before posting.
- If validation fails, the journal must not be posted.

2.6. Custom / Field Mapping

Field	Source	D365 Data field Name	Comments
Name	D365	Gen Jrn	Select from the batch job parameter
Journal	D365		Use the OOTB behavior of D365 to create a journal number

Line Record

Field	Data Source	D365 Data field Name	Comments
Date	LedgerJournalTrans	TransDate	Date set in the Batch Job parameter
Account type	LedgerJournalTrans	AccountType	Account type for transaction i.e Project
Account	LedgerJournalTrans	LedgerDimension	Project ID

Debit	LedgerJournalTrans	AmountCurDebit	Amount
Credit	LedgerJournalTrans	AmountCurCredit	Amount
Offset account type	LedgerJournalTrans	OffsetAccountType	Account type for transaction i.e Ledger
Offset account	LedgerJournalTrans	OffsetLedgerDimension	Main account and dimension from Direct Labor GL Journal (paycom GL account default dimension)
Activity Number	LedgerJournalTrans	ActivityNumber	Activity number from AgriERP consumption journal
Category	LedgerJournalTrans	CategoryId	Use the OOTB behavior of D365
Resource	LedgerJournalTrans	Resource	Resource from AgriERP consumption journal
Currency	LedgerJournalTrans		Use the OOTB behavior of D365
Line Property	LedgerJournalTrans	ProjLinePropertyId	Use the OOTB behavior of D365
Financial Dimension	D365		Use the custom behavior as described in other FDDs

Activity Number Logic:

The Activity number on each project transaction is not entered manually; it is derived from the project's Work Breakdown Structure (WBS) as follows:

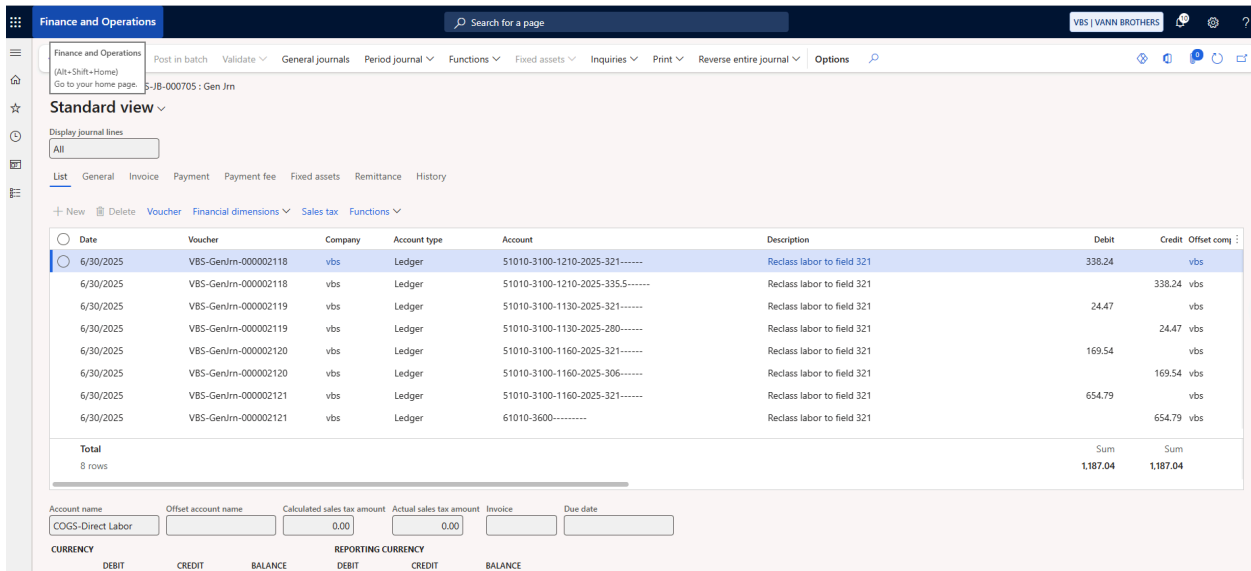
- 1- The Farming Activity is read from the Absorption account posting line of the transaction.
- 2- Each WBS line of the project carries an Operation ID, and the Farming Activity is the same as that Operation ID.
- 3- The WBS line whose Operation ID matches the Farming Activity is mapped to a specific Activity number.
- 4- That mapped Activity number is picked and applied as the Activity number on the transaction.

Absorption account posting line → Farming Activity → matched to the project WBS Operation ID → the Activity number mapped on that WBS line.

For Cost Center 3510:

Overhead (General Farm) cost center, the Absorption account posting line does not carry a Farming Activity. Since the standard derivation relies on a Farming Activity being present on the absorption line, it cannot be applied to this cost center, so the following exception applies:

- 1 When the cost center on the transaction is 3510 — Overhead (General Farm) — i.e., the absorption account posting line has no Farming Activity associated — the Operation ID is defaulted (fixed) to 1720 — Farm Overhead
- 2 The Activity number is then picked from the project's WBS line whose Operation ID is 1720 (Farm Overhead).



The screenshot displays the 'Finance and Operations' module with a search bar and navigation tabs. The 'Standard view' is selected, showing a list of journal entries. The table below represents the data shown in the screenshot.

Date	Voucher	Company	Account type	Account	Description	Debit	Credit	Offset com
6/30/2025	VBS-GenJrn-000002118	vbs	Ledger	51010-3100-1210-2025-321-----	Reclass labor to field 321	338.24		vbs
6/30/2025	VBS-GenJrn-000002118	vbs	Ledger	51010-3100-1210-2025-335.5-----	Reclass labor to field 321		338.24	vbs
6/30/2025	VBS-GenJrn-000002119	vbs	Ledger	51010-3100-1130-2025-321-----	Reclass labor to field 321	24.47		vbs
6/30/2025	VBS-GenJrn-000002119	vbs	Ledger	51010-3100-1130-2025-280-----	Reclass labor to field 321		24.47	vbs
6/30/2025	VBS-GenJrn-000002120	vbs	Ledger	51010-3100-1160-2025-321-----	Reclass labor to field 321	169.54		vbs
6/30/2025	VBS-GenJrn-000002120	vbs	Ledger	51010-3100-1160-2025-306-----	Reclass labor to field 321		169.54	vbs
6/30/2025	VBS-GenJrn-000002121	vbs	Ledger	51010-3100-1160-2025-321-----	Reclass labor to field 321	654.79		vbs
6/30/2025	VBS-GenJrn-000002121	vbs	Ledger	61010-3600-----	Reclass labor to field 321		654.79	vbs
Total						Sum	Sum	
8 rows						1,187.04	1,187.04	

Below the table, there are input fields for 'Account name' (COGS-Direct Labor), 'Offset account name', 'Calculated sales tax amount' (0.00), 'Actual sales tax amount' (0.00), 'Invoice', and 'Due date'. At the bottom, there are tabs for 'CURRENCY' (DEBIT, CREDIT, BALANCE) and 'REPORTING CURRENCY' (DEBIT, CREDIT, BALANCE).

Default Dimension Assignment

As per our customization, Dimension will be derived from master data into projects and Farming Activity will be derived from Activity ID of the project.

2.7. Other Requirements

Processing Requirements

- Only posted transactions will be considered.
- Transactions must fall within the selected date range.
- GL transactions must be limited to the Labor absorption account, e.g., 51011.
- Project Expense transactions must be linked to a Project that has a Plot/Field reference.
- The process must prevent duplicate variance posting for the same period and Plot/Field.

Error Handling

- Missing setup accounts must stop the process.
- Missing Plot/Field on GL transactions must be logged.
- Missing Plot/Field mapping on the Project must be logged.
- Journal validation errors must be displayed to the user